

ON REGRESSION AND SCEDASTIC EQUATIONS

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1. Introduction

Using the moment generating function and its derivatives as starting point, Steyn (1957, 1960) derived and applied a necessary and sufficient condition for rational regression equations. In this paper a similar approach is used to derive a necessary and sufficient condition for prescribed forms of the scedastic equations. The cases considered here are the following: (i) Given the rational regression equation of one random variable on the remaining $p-1$ random variables, a necessary and sufficient condition for the scedastic equation to be rational is derived; (ii) Using logical extensions in notations, the theorems are generalized to include random vectors and random matrices. Results derived by Mathai (1967) are used and extended.

The theorems are applied to a few multivariate distributions.

2. Regression and Scedasticity of Random Variables

Consider p random variables $X_1, X_2, \dots, X_p$ with probability density function $f(x_1, x_2, \dots, x_p)$, $a_i \leq x_i \leq b_i$, $i=1, 2, \dots, p$ and moment generating function

$$M(a_1, a_2, \dots, a_p) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \dots \int_{a_p}^{b_p} \exp\left(\sum_{i=1}^p a_i x_i\right) f(x_1, x_2, \dots, x_p) \prod_{i=1}^p dx_i.$$

where a_i , $i=1, 2, \dots, p$ are complex.

Steyn (1957) proved the following theorem:

Theorem 2.1

Given p random variables $X_1, X_2, \dots, X_p$, then a necessary and sufficient condition that the regression equation of X_1 on $X_2, X_3, \dots, X_p$ is given by

$$E(X_1 | X_2 = x_2, \dots, X_p = x_p) = \frac{Q(x_2, \dots, x_p)}{R(x_2, \dots, x_p)},$$

where Q and R are polynomials, is

$$(2.1) \quad \left[\frac{\partial}{\partial a_1} R \left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p} \right) M = Q \left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p} \right) M \right]_{a=0},$$

where $M = M(a_1, a_2, \dots, a_p)$ is the moment generating function of $X_1, X_2, \dots, X_p$ and

$$\left(\frac{\partial}{\partial a_i} \right)^r = \frac{\partial^r}{\partial a_i^r} \quad \text{for } i=1, 2, \dots, p.$$

The following theorem will now be proved:

Theorem 2.2

Given p random variables $X_1, X_2, \dots, X_p$ with rational regression equation

$$(2.2) \quad E(X_1 | X_2 = x_2, \dots, X_p = x_p) = \frac{Q(x_2, \dots, x_p)}{R(x_2, \dots, x_p)}$$

then a necessary and sufficient condition that the scedastic equation of X_1 on $X_2, X_3, \dots, X_p$ is given by

$$(2.3) \quad V(X_1 | X_2 = x_2, \dots, X_p = x_p) = \frac{L(x_2, \dots, x_p)}{N(x_2, \dots, x_p)}$$

where Q, R, L and N are polynomials in $x_2, \dots, x_p$, follows as

$$\begin{aligned} & \left[\frac{\partial^2}{\partial a_1^2} \left[R \left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p} \right) \right]^2 \left[N \left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p} \right) \right]^M \left[Q \left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p} \right) \right]^2 \left[N \left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p} \right) \right]^M \right. \\ & \quad \left. = \left[R \left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p} \right) \right]^2 \left[L \left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p} \right) \right]^M \right]_{a_1=0}, \end{aligned}$$

where $\left(\frac{\partial}{\partial a_i} \right)^r \left(\frac{\partial}{\partial a_j} \right)^s = \frac{\partial^{r+s}}{\partial a_i^r \partial a_j^s}$

Proof

When (2.2) and (2.3) are given it follows that,

$$(2.4) \quad \frac{\int_{a_1}^{b_1} x_1 f(x_1, x_2, \dots, x_p) dx_1}{\int_{a_1}^{b_1} f(x_1, x_2, \dots, x_p) dx_1} = \frac{Q(x_2, \dots, x_p)}{R(x_2, \dots, x_p)}$$

and

$$(2.5) \quad \frac{\int_{a_1}^{b_1} \left(x_1 - \frac{Q(x_2, \dots, x_p)}{R(x_2, \dots, x_p)} \right)^2 f(x_1, x_2, \dots, x_p) dx_1}{\int_{a_1}^{b_1} f(x_1, x_2, \dots, x_p) dx_1} = \frac{L(x_2, \dots, x_p)}{N(x_2, \dots, x_p)}.$$

By using (2.4), (2.5) can be written as

$$\frac{\int_{a_1}^{b_1} x_1^2 f(x_1, x_2, \dots, x_p) dx_1}{\int_{a_1}^{b_1} f(x_1, x_2, \dots, x_p) dx_1} - \left[\frac{Q(x_2, \dots, x_p)}{R(x_2, \dots, x_p)} \right]^2 = \frac{L(x_2, \dots, x_p)}{N(x_2, \dots, x_p)}$$

so that

$$\begin{aligned}
& [R(x_2, \dots, x_p)]^2 [N(x_2, \dots, x_p)] \int_{a_1}^{b_1} x_1^2 f(x_1, x_2, \dots, x_p) dx_1 \\
& - [Q(x_2, \dots, x_p)]^2 [N(x_2, \dots, x_p)] \int_{a_1}^{b_1} f(x_1, x_2, \dots, x_p) dx_1 \\
& = [R(x_2, \dots, x_p)]^2 [L(x_2, \dots, x_p)] \int_{a_1}^{b_1} f(x_1, x_2, \dots, x_p) dx_1
\end{aligned}$$

Multiplying by $\exp(\sum_{i=2}^p a_i x_i)$ and integrating with respect to $x_1, x_2, \dots, x_p$, it follows that

$$\begin{aligned}
(2.6) \quad & \left[\frac{\partial^2}{\partial a_1^2} \left[R\left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p}\right) \right]^2 \left[N\left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p}\right) \right] M \cdot \left[Q\left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p}\right) \right]^2 \left[N\left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p}\right) \right] M \right. \\
& \left. = \left[R\left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p}\right) \right]^2 \left[L\left(\frac{\partial}{\partial a_2}, \dots, \frac{\partial}{\partial a_p}\right) \right] M \right]_{a_1=0},
\end{aligned}$$

where the moment generating function $M(a_1, a_2, \dots, a_p)$ is assumed to exist for an $a_1, a_2, \dots, a_p$ region, including $a_1=0$. Thus equation (2.6) provides a necessary condition for a rational scedastic equation of X_1 on $X_2, X_3, \dots, X_p$.

That (2.6) is a sufficient condition as well, can be shown as follows:

Given (2.6) and (2.2) it follows that

$$\begin{aligned}
(2.7) \quad & \int_{a_2}^{b_2} \dots \int_{a_p}^{b_p} \exp\left(\sum_{i=2}^p a_i x_i\right) [R(x_2, \dots, x_p)]^2 [N(x_2, \dots, x_p)] \\
& \cdot \int_{a_1}^{b_1} (x_1 - E(X_1 | X_2 = x_2, \dots, X_p = x_p))^2 f(x_1, x_2, \dots, x_p) dx_1 \prod_{i=2}^p dx_i \\
& = \int_{a_2}^{b_2} \int_{a_p}^{b_p} \exp\left(\sum_{i=2}^p a_i x_i\right) [R(x_2, \dots, x_p)]^2 [L(x_2, \dots, x_p)] \cdot \int_{a_1}^{b_1} f(x_1, x_2, \dots, x_p) dx_1 \prod_{i=2}^p dx_i
\end{aligned}$$

The function in $a_2, \dots, a_p$ on the left hand side of (2.7) is a generalized Laplace transform of

$$\begin{aligned}
(2.8) \quad & [R(x_2, \dots, x_p)]^2 [N(x_2, \dots, x_p)] \int_{a_1}^{b_1} (x_1 - E(X_1 | X_2 = x_2, \dots, X_p = x_p))^2 \\
& \cdot f(x_1, x_2, \dots, x_p) dx_1
\end{aligned}$$

while the right hand side of (2.7) is a generalized Laplace transform of

$$(2.9) \quad [R(x_2, \dots, x_p)]^2 [L(x_2, \dots, x_p)] \int_{a_1}^{b_1} f(x_1, x_2, \dots, x_p) dx_1.$$

Hence, it follows from the Uniqueness theorem of Laplace transforms (Widder (1941)) that, under certain general conditions, the two functions in (2.8) and (2.9) are the same, so that

$$V(X_1 | X_2 = x_2, \dots, X_p = x_p) = \frac{L(x_2, \dots, x_p)}{N(x_2, \dots, x_p)}$$

This proves the theorem.

Similar conditions hold for the regression equations and scedastic equation of any X_i , $i = 1, 2, \dots, p$ on the remaining variables.

The above theorem was proved for the continuous case. It is immediately clear that the theorem also holds for the discrete case by using summations instead of integrations. Further, when the probability generating function $G(t_1, t_2, \dots, t_p)$ is used as starting point, it follows immediately from $t_i = e^{a_i}$, that the necessary and sufficient conditions (2.1) and (2.6) may be written down by merely substituting $G(t_1, t_2, \dots, t_p)$ for $M(a_1, a_2, \dots, a_p)$, replacing $\frac{\partial}{\partial a_i}$ by $t_i \frac{\partial}{\partial t_i}$ and remembering that the operator

$$(2.10) \quad (t_i \frac{\partial}{\partial t_i})^r = (t_i \frac{\partial}{\partial t_i}) (t_i \frac{\partial}{\partial t_i}) \dots (t_i \frac{\partial}{\partial t_i}),$$

i.e. applying the operator $t_i \frac{\partial}{\partial t_i}$ r times, while the condition $a_i = 0$ becomes $t_i = 1$.

The necessary and sufficient condition (2.6) may seem complex, but it will be clear from the examples below that the application is straight-forward.

Example 2.1

The bivariate normal distribution with zero means (Wilks (1962)) has a moment generating function:

$$M = M(a_1, a_2) = \exp\left(\frac{1}{2}(\sigma_1^2 a_1^2 + 2\rho\sigma_1\sigma_2 a_1 a_2 + \sigma_2^2 a_2^2)\right)$$

Clearly,

$$(2.11) \quad \begin{cases} \frac{\partial}{\partial a_1} M = (\sigma_1^2 a_1 + \rho\sigma_1\sigma_2 a_2)M \\ \frac{\partial}{\partial a_2} M = (\sigma_2^2 a_2 + \rho\sigma_1\sigma_2 a_1)M \end{cases}$$

$$(2.12) \quad \frac{\partial^2}{\partial a_1^2} M = (\sigma_1^2 a_1 + \rho\sigma_1\sigma_2 a_2)^2 M + \sigma_1^2 M, \text{ and}$$

$$(2.13) \quad \frac{\partial^2}{\partial a_2^2} M = (\sigma_2^2 a_2 + \rho\sigma_1\sigma_2 a_1)^2 M + \sigma_2^2 M.$$

Putting $a_1 = 0$ in (2.11) then

$$\left[\frac{\partial}{\partial a_1} M + \rho \frac{\sigma_1}{\sigma_2} \frac{\partial}{\partial a_2} M \right]_{a_1=0} = 0$$

and by applying theorem 2.1 it follows that

$$E(X_1 | X_2 = x_2) = \rho \frac{\sigma_1}{\sigma_2} x_2.$$

Also from (2.12) and (2.13) with $a_1 = 0$ follows,

$$\left[\frac{\partial^2}{\partial a_1^2} M + \rho^2 \frac{\sigma_1^2}{\sigma_2^2} \frac{\partial^2}{\partial a_2^2} M = \sigma_1^2 (1 - \rho^2) M \right]_{a_1=0}$$

so that by theorem 2.2,

$$V(X_1 | X_2 = x_2) = \sigma_1^2 (1 - \rho^2),$$

as is well known.

Example 2.2

The Dirichlet probability function of the first kind is given by (Wilks (1962))

$$(2.14) f \approx f(x_1, x_2, \dots, x_p) = \frac{\Gamma(\frac{1}{2}m + \frac{1}{2}n)}{\prod_{j=1}^p \Gamma(\frac{1}{2}m_j) \Gamma(\frac{1}{2}n)} \prod_{j=1}^p x_j^{\frac{1}{2}m_j - 1} (1 - \sum_{j=1}^p x_j)^{\frac{1}{2}n - 1}$$

where $x_j > 0$, $1 - \sum_{j=1}^p x_j > 0$ and $m = \sum_{j=1}^p m_j$.

Since the moment generating function of this distribution is not well known it is preferred to use the probability function as starting point.

Differentiating (2.14) with respect to x_1 it follows that

$$x_1 (1 - \sum_{j=1}^p x_j) \frac{\partial}{\partial x_1} f = [(\frac{1}{2}m_1 - 1)(1 - \sum_{j=1}^p x_j) - (\frac{1}{2}n - 1)x_1] f,$$

so that in terms of the moment generating function (Steyn (1960) eq. (6), (7) it follows

$$(2.15) \quad -\frac{\partial}{\partial a_1} (a_1 M) + \sum_{j=1}^p \frac{\partial}{\partial a_1} \frac{\partial}{\partial a_j} (a_1 M) = (\frac{1}{2}m_1 - 1) M - (\frac{1}{2}m_1 - 1) \sum_{j=1}^p \frac{\partial}{\partial a_j} M - (\frac{1}{2}n - 1) \frac{\partial}{\partial a_1} M.$$

After completion of the differentiation in (2.15) it follows from theorem 2.1 that

$$E(X_1 | X_2 = x_2, \dots, X_p = x_p) = \frac{m_1}{n + m_1} (1 - \sum_{j=2}^p x_j).$$

Differentiate (2.15) again with respect to a_1 ; then

$$-\frac{\partial^2}{\partial a_1^2} (a_1 M) + \sum_{j=1}^p \frac{\partial^2}{\partial a_1^2} \frac{\partial}{\partial a_j} (a_1 M) = (\frac{1}{2}m_1 - 1) \frac{\partial}{\partial a_1} M - (\frac{1}{2}m_1 - 1) \sum_{j=1}^p \frac{\partial^2}{\partial a_1 \partial a_j} M - (\frac{1}{2}n - 1) \frac{\partial^2}{\partial a_1^2} M.$$

Substituting for $\frac{\partial}{\partial a_1} M$ from (2.15) and putting $a_1 = 0$ the above equation becomes

$$\left[\frac{\partial^2}{\partial a_1^2} M \cdot \frac{m_1^2}{(n + m_1)^2} \left(1 - \sum_{j=2}^p \frac{\partial}{\partial a_j}\right)^2 M + \frac{2n m_1}{(n + m_1)^2 (n + m_1 + 2)} \left(1 - \sum_{j=2}^p \frac{\partial}{\partial a_j}\right)^2 M \right]_{a_1=0}$$

Hence:

$$V(X_1 | X_2 = x_2, \dots, X_p = x_p) = \frac{2n m_1}{(n + m_1)^2 (n + m_1 + 2)} \left(1 - \sum_{j=2}^p x_j\right)^2 \\ = \frac{2n}{m_1 (n + m_1 + 2)} [E(X_1 | X_2 = x_2, \dots, X_p = x_p)]^2$$

Example 2.3

The bivariate Poissonian probability (correlation) function (Aitken (1952)) has probability generating function:

$$G = G(t_1, t_2) = \exp [\mu(t_1 - 1) + \mu'(t_2 - 1) + \bar{\mu}(t_1 - 1)(t_2 - 1)]$$

Clearly

$$\frac{\partial}{\partial t_1} G = (\mu - \bar{\mu} + \bar{\mu} t_2) G, \quad \frac{\partial}{\partial t_2} G = (\mu' - \bar{\mu}' + \bar{\mu}' t_1) G$$

and

$$\frac{\partial^2}{\partial t_1^2} G = (\mu - \bar{\mu} + \bar{\mu} t_2) \frac{\partial}{\partial t_1} G, \quad \frac{\partial^2}{\partial t_1^2} G = (\mu' - \bar{\mu}' + \bar{\mu}' t_1) \frac{\partial}{\partial t_2} G$$

Thus,

$$\left[t_1 \frac{\partial}{\partial t_1} G = (\mu - \bar{\mu} + \frac{\bar{\mu}}{\mu'} t_2 \frac{\partial}{\partial t_2}) G \right]_{t_1 = 1}$$

giving

$$E(X_1 | X_2 = x_2) = \mu - \bar{\mu} + \frac{\bar{\mu}}{\mu'} x_2.$$

Also from

$$(t_1 \frac{\partial}{\partial t_1})^2 G = t_1^2 \frac{\partial^2}{\partial t_1^2} G + t_1 \frac{\partial}{\partial t_1} G,$$

follows for $t_1 = 1$ that,

$$\left[(t_1 \frac{\partial}{\partial t_1})^2 G - (\mu - \bar{\mu} + \frac{\bar{\mu}}{\mu'} t_2 \frac{\partial}{\partial t_2})^2 G = (\mu - \bar{\mu} + \frac{\bar{\mu}}{\mu'} (1 - \frac{\bar{\mu}}{\mu'}) t_2 \frac{\partial}{\partial t_2}) G \right]_{t_1 = 1}$$

and

$$V(X_1 | X_2 = x_2) = \mu - \bar{\mu} + \frac{\bar{\mu}}{\mu'} (1 - \frac{\mu}{\mu'}) x_2$$

3. Regression and Scedasticity of Random Vectors

Consider p random vectors $X^{(1)}, X^{(2)}, \dots, X^{(p)}$ with probability density function $f(x^{(1)}, x^{(2)}, \dots, x^{(p)})$ and moment generating function

$$M(a^{(1)}, a^{(2)}, \dots, a^{(p)}) = E[\exp(\sum_{i=1}^p x^{(i)} a^{(i)})]$$

$$= \int \dots \int \exp(\sum_{i=1}^p x^{(i)} a^{(i)}) f(x^{(1)}, x^{(2)}, \dots, x^{(p)}) \prod_{i=1}^p dx^{(i)}$$

Mathai (1967) derived a necessary and sufficient condition for linear regression of one random vector $X^{(1)}$ on vectors $X^{(2)}, \dots, X^{(p)}$. The following theorem is a generalization of Mathai's theorem.

Theorem 3.1 Given p random column vectors $X^{(1)}, X^{(2)}, \dots, X^{(p)}$ with finite expected values, then a necessary and sufficient condition that the regression equation of $X^{(1)}$ on $X^{(2)}, \dots, X^{(p)}$ is given by

$$(3.1) E(X^{(1)} | X^{(2)} = x^{(2)}, \dots, X^{(p)} = x^{(p)}) R(x^{(2)}, \dots, x^{(p)}) = Q(x^{(2)}, \dots, x^{(p)})$$

where R and Q are such polynomials that both sides of (3.1) are vectors or matrices respectively, i.e. R must be either a row vector with the same number of elements as $X^{(1)}$ or a scalar while Q must be either a square matrix or a column vector, is

$$\left[\frac{\partial}{\partial a^{(1)}} R \left(\frac{\partial}{\partial a^{(2)}}, \dots, \frac{\partial}{\partial a^{(p)}} \right) M(a^{(1)}, \dots, a^{(p)}) = Q \left(\frac{\partial}{\partial a^{(2)}}, \dots, \frac{\partial}{\partial a^{(p)}} \right) M(a^{(1)}, \dots, a^{(p)}) \right]_{a^{(1)}=0}$$

Proof

The proof is similar to that given by Mathai and will only be outlined briefly. From (3.1) follows on multiplying both sides by

$$\exp \left(\sum_{i=2}^p X^{(i)'} a^{(i)} \right)$$

and taking the mathematical expectations, that

$$E [X^{(1)} R(X^{(2)}, \dots, X^{(p)}) \exp \left(\sum_{i=2}^p X^{(i)'} a^{(i)} \right)] = E [Q(X^{(2)}, \dots, X^{(p)}) \exp \left(\sum_{i=2}^p X^{(i)'} a^{(i)} \right)] \quad (3.2)$$

since

$$E(E(X^{(1)} | X^{(2)} = x^{(2)}, \dots, X^{(p)} = x^{(p)})) = E(X^{(1)}).$$

Next,

$$\left[\frac{\partial}{\partial a^{(1)}} M(a^{(1)}, \dots, a^{(p)}) \right]_{a^{(1)}=0} = \left[E \left(\frac{\partial}{\partial a^{(1)}} \exp \left(\sum_{i=1}^p X^{(i)'} a^{(i)} \right) \right) \right]_{a^{(1)}=0}$$

$$= E [X^{(1)} \exp \left(\sum_{i=2}^p X^{(i)'} a^{(i)} \right)] \quad \text{and}$$

$$\left[\frac{\partial}{\partial a^{(i)}} M(a^{(1)}, \dots, a^{(p)}) \right]_{a^{(1)}=0} = E [X^{(i)} \exp \left(\sum_{i=2}^p X^{(i)'} a^{(i)} \right)] \quad \text{for } i=2, \dots, p.$$

Equation (3.1) thus leads to

$$\left[\frac{\partial}{\partial a^{(1)}} R \left(\frac{\partial}{\partial a^{(2)}}, \dots, \frac{\partial}{\partial a^{(p)}} \right) M(a^{(1)}, \dots, a^{(p)}) - Q \left(\frac{\partial}{\partial a^{(2)}}, \dots, \frac{\partial}{\partial a^{(p)}} \right) M(a^{(1)}, \dots, a^{(p)}) \right]_{a^{(1)}=0}$$

This proves the necessary condition. That the condition is also sufficient is proved as follows. Given (3.3) then (3.2) follows. Thus

$$E \exp \left(\sum_{i=1}^p X^{(i)'} a^{(i)} \right) [E [(X^{(1)} R(X^{(2)}, \dots, X^{(p)}) - Q(X^{(2)}, \dots, X^{(p)})) | X^{(2)} = x^{(2)}, \dots, X^{(p)} = x^{(p)}]] = 0 \quad (3.4)$$

where 0 is a null vector. Considering (3.4) as a generalized Laplace transform it follows under certain general conditions, that

$$E [X^{(1)} R(X^{(2)}, \dots, X^{(p)}) - Q(X^{(2)}, \dots, X^{(p)}) | X^{(2)} = x^{(2)}, \dots, X^{(p)} = x^{(p)}] = 0$$

Thus

$$E(X^{(1)} | X^{(2)}=x^{(2)}, \dots, X^{(p)}=x^{(p)}) R(x^{(2)}, \dots, x^{(p)}) = Q(x^{(2)}, \dots, x^{(p)}).$$

This proves the theorem.

A theorem similar to theorem 2.2 will now be stated for random vectors.

Theorem 3.2

Given p random vectors $X^{(1)}, X^{(2)}, \dots, X^{(p)}$ with finite expected values and regression equation of $X^{(1)}$ on $X^{(2)}, X^{(3)}, \dots, X^{(p)}$,

$$E(X^{(1)} | X^{(2)}=x^{(2)}, \dots, X^{(p)}=x^{(p)}) R(x^{(2)}, \dots, x^{(p)}) = Q(x^{(2)}, \dots, x^{(p)})$$

then a necessary and sufficient condition that the scedastic equation of $X^{(1)}$ on $X^{(2)}, X^{(3)}, \dots, X^{(p)}$ is given by

$$\begin{aligned} & V(X^{(1)} | X^{(2)}=x^{(2)}, \dots, X^{(p)}=x^{(p)}) N(x^{(2)}, \dots, x^{(p)}) \\ & \equiv E [X^{(1)} - E(X^{(1)} | X^{(2)}=x^{(2)}, \dots, X^{(p)}=x^{(p)})] (X^{(1)} - E(X^{(1)} | X^{(2)}=x^{(2)}, \dots, X^{(p)}=x^{(p)}))' \\ (3.5) \quad & - [X^{(2)}=x^{(2)}, \dots, X^{(p)}=x^{(p)}] N(x^{(2)}, \dots, x^{(p)}) \\ & = L(x^{(2)}, \dots, x^{(p)}). \end{aligned}$$

where N and L are square matrix polynomials, follows as

$$\begin{aligned} & \left[\left(\frac{\partial}{\partial a^{(1)}} \right)' \left(\frac{\partial}{\partial a^{(2)}} \right)' \dots \left(\frac{\partial}{\partial a^{(p)}} \right)' \right] \left[R \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right]' \left[R \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right] \left[N \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right] \\ & \cdot M \cdot \left[Q \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right] \left[Q \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right]' \left[N \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right] M \\ & = \left[R \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right]' \left[R \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right] \left[L \left(\frac{\partial}{\partial a^{(2)}} \right), \dots, \frac{\partial}{\partial a^{(p)}} \right] M \Big|_{a^{(1)}=0} \end{aligned}$$

where $M \equiv M(a^{(1)}, a^{(2)}, \dots, a^{(p)})$ the moment generating function of $X^{(1)}, X^{(2)}, \dots, X^{(p)}$ and $\left(\frac{\partial}{\partial a^{(k)}} \right)' \left(\frac{\partial}{\partial a^{(l)}} \right)' = \left(\frac{\partial^2}{\partial a_k \partial a_l} \right)$ for $k, l = 1, 2, \dots, p$.

The proof of this theorem for random vectors is similar to the proof of theorem 2.2 and of theorem 4.1 for random matrices, which will also be proved fully. If R is a scalar then $R'R$ becomes R^2 .

The scedastic equations for most of the classical multivariate distributions follow easily by straightforward differentiation and application of theorem 3.2. The multivariate normal and multivariate Poisson are used as examples.

Example 3.1

The multivariate normal distribution (Anderson (1958)) has a moment generating function

$$(3.6) \quad M \equiv M(a^{(1)}, a^{(2)}) = \exp \left(\frac{1}{2} a^{(1)'} \Sigma_{11} a^{(1)} + 2 a^{(2)'} \Sigma_{21} a^{(1)} + a^{(2)'} \Sigma_{22} a^{(2)} \right)$$

$$\text{where } \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

From (3.6) follows on differentiation that:

$$(3.7) \quad \frac{\partial}{\partial a^{(1)}} M = (\Sigma_{11} a^{(1)} + \Sigma_{12} a^{(2)}) M,$$

$$(3.8) \quad \frac{\partial}{\partial a^{(2)}} M = (\Sigma_{22} a^{(2)} + \Sigma_{21} a^{(1)}) M$$

and

$$(3.9) \quad \left(\frac{\partial}{\partial a^{(1)}} \right) \left(\frac{\partial}{\partial a^{(1)}} \right)' M = (\Sigma_{11} a^{(1)} + \Sigma_{12} a^{(2)}) (\Sigma_{11} a^{(1)} + \Sigma_{12} a^{(2)})' M + \Sigma_{11} M,$$

$$(3.10) \quad \left(\frac{\partial}{\partial a^{(2)}} \right) \left(\frac{\partial}{\partial a^{(2)}} \right)' M = (\Sigma_{22} a^{(2)} + \Sigma_{21} a^{(1)}) (\Sigma_{22} a^{(2)} + \Sigma_{21} a^{(1)})' M + \Sigma_{22} M.$$

Putting $a^{(1)}=0$ in (3.7) and (3.8) then

$$\left[\frac{\partial}{\partial a^{(1)}} M = \Sigma_{12} \Sigma_{22}^{-1} \frac{\partial}{\partial a^{(2)}} M \right]_{a^{(1)}=0}$$

By applying theorem 3.1, it follows that

$$E(X^{(1)} | X^{(2)} = x^{(2)}) = \Sigma_{12} \Sigma_{22}^{-1} x^{(2)}.$$

Also from (3.9) and (3.10) with $a^{(1)}=0$ follows,

$$\left[\left(\frac{\partial}{\partial a^{(1)}} \right) \left(\frac{\partial}{\partial a^{(1)}} \right)' M - (\Sigma_{12} \Sigma_{22}^{-1} \frac{\partial}{\partial a^{(2)}}) (\Sigma_{12} \Sigma_{22}^{-1} \frac{\partial}{\partial a^{(2)}})' M = \Sigma_{11.2} M \right]_{a^{(1)}=0}$$

where $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$, so that by theorem 3.2

$$V(X^{(1)} | X^{(2)} = x^{(2)}) = \Sigma_{11.2}$$

as is well known.

Example 3.2

A generalized Poisson correlation function is obtained from the probability generating function:

$$(3.11) \quad G = G(t^{(1)}, t^{(2)}) = \exp\{ (t^{(1)} - \underline{1})' \mu^{(1)} + (t^{(2)} - \underline{1})' \mu^{(2)} + (t^{(1)} - \underline{1})' \bar{\mu}^{(2)} - \underline{1} \}$$

$$\text{where } t^{(1)} = \begin{pmatrix} t_1 \\ t_2 \\ \vdots \\ t_r \end{pmatrix}, t^{(2)} = \begin{pmatrix} t_{r+1} \\ t_{r+2} \\ \vdots \\ t_p \end{pmatrix}, \underline{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix},$$

$$\mu^{(1)} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_r \end{pmatrix}, \mu^{(2)} = \begin{pmatrix} \mu_{r+1} \\ \mu_{r+2} \\ \vdots \\ \mu_p \end{pmatrix} \quad \text{and } \bar{\mu} = [\bar{\mu}_{ij}] \quad , i=1, \dots, r, j=r+1, \dots, p,$$

an $r \times (p-r)$ matrix. The probability of function $f(X^{(1)}, X^{(2)})$,

$$X^{(1)} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_r \end{pmatrix} \text{ and } X^{(2)} = \begin{pmatrix} X_{r+1} \\ X_{r+2} \\ \vdots \\ X_p \end{pmatrix},$$

can take all values given by the coefficient of $\prod_{i=1}^p t_i^{x_i}$ in the expansion of $G(t^{(1)}, t^{(2)})$. This interesting multivariate distribution is obtained for example when two sets of rare events are considered under the condition that one event in each set may occur in the same trial but events in the same set are mutually exclusive.

From (3.11) follows on differentiation that:

$$(3.12) \quad \frac{\partial}{\partial t^{(1)}} G = (\mu^{(1)} + \bar{\mu}(t^{(2)} - \underline{1})) G,$$

$$(3.13) \quad \frac{\partial}{\partial t^{(2)}} G = (\mu^{(2)} + \bar{\mu}'(t^{(1)} - \underline{1})) G \text{ and}$$

$$(3.14) \quad \left(-\frac{\partial}{\partial t^{(1)}}\right) \left(-\frac{\partial}{\partial t^{(1)}}\right)' G = -\frac{\partial}{\partial t^{(1)}} G (\mu^{(1)} + \bar{\mu}(t^{(2)} - \underline{1}))'.$$

Putting $t^{(1)} = \underline{1}$ in (3.12) and (3.13) so that $t_i \frac{\partial}{\partial t_i} = \frac{\partial}{\partial t_i}$, $i=1,2,\dots,r$ and substituting $\frac{\partial}{\partial t_j} G = \mu_j G$, $j=r+1,\dots,p$ in the elements of the vector on the righthand side of (3.12), it follows immediately that

$$E(X^{(1)} | X^{(2)} = x^{(2)}) = \mu^*(1) + \bar{\mu}^* x^{(2)} \text{ where}$$

$$\mu^*(1) = \begin{pmatrix} \mu_1 - \sum_{j=r+1}^p \bar{\mu}_{1j} \\ \mu_2 - \sum_{j=r+1}^p \bar{\mu}_{2j} \\ \vdots \\ \mu_r - \sum_{j=r+1}^p \bar{\mu}_{rj} \end{pmatrix} \text{ and } \bar{\mu}^* = \begin{bmatrix} \bar{\mu}_{ij} \\ \mu_j \end{bmatrix}$$

Since for $t^{(1)} = \underline{1}$,

$$\begin{pmatrix} t_1 \frac{\partial}{\partial t_1} \\ t_2 \frac{\partial}{\partial t_2} \\ \vdots \\ t_r \frac{\partial}{\partial t_r} \end{pmatrix} (t_1 \frac{\partial}{\partial t_1}, t_2 \frac{\partial}{\partial t_2}, \dots, t_r \frac{\partial}{\partial t_r}) G = \left(-\frac{\partial}{\partial t^{(1)}}\right) \left(-\frac{\partial}{\partial t^{(1)}}\right)' G + [\delta_{ij} \frac{\partial}{\partial t_i}] G$$

where $\delta_{ij} = 1$ ($i=j$)
 $= 0$ ($i \neq j$),

it follows from (3.14) and (2.10) that

$$V(X^{(1)} | X^{(2)} = x^{(2)}) = [\delta_{ij} \quad \lambda_i]$$

$$\text{where } \lambda_i = \mu_i - \sum_{j=r+1}^p \mu_{ij} - \sum_{j=r+1}^p \frac{\bar{\mu}_{ij}}{\bar{\mu}_j} (1 - \frac{\bar{\mu}_{ij}}{\bar{\mu}_j}) x_j$$

4. Regression and Scedasticity for Random Matrices

Consider p random matrices $X_{(1)}, X_{(2)}, \dots, X_{(p)}$ with probability density function $f(x_{(1)}, x_{(2)}, \dots, x_{(p)})$ and moment generating function

$$\begin{aligned} M(a_{(1)}, a_{(2)}, \dots, a_{(p)}) &= E [\exp (\operatorname{tr} \sum_{i=1}^p a_{(i)} X'_{(i)})] \\ &= \int \dots \int \exp (\operatorname{tr} \sum_{i=1}^p a_{(i)} X'_{(i)}) f(x_{(1)}, x_{(2)}, \dots, x_{(p)}) \prod_{i=1}^p dx_{(i)}. \end{aligned}$$

According to Mathai (1967), the following necessary and sufficient condition for rational regression in the case of random symmetric matrices may be written down as

$$\left[\frac{\partial}{\partial a_{(1)}} R \left(\frac{\partial}{\partial a_{(2)}}, \dots, \frac{\partial}{\partial a_{(p)}} \right) M(a_{(1)}, \dots, a_{(p)}) - Q \left(\frac{\partial}{\partial a_{(2)}}, \dots, \frac{\partial}{\partial a_{(p)}} \right) M(a_{(1)}, \dots, a_{(p)}) \right]_{a_{(1)}=0}$$

where Q and R are polynomials in $x_{(2)}, x_{(3)}, \dots, x_{(p)}$.

The above theorem holds also for unsymmetric matrices.

The following theorem will now be proved:

Theorem 4.1

Given p square random matrices $X_{(1)}, X_{(2)}, \dots, X_{(p)}$ of the same finite order. Let

$$E(X_{(1)}, X_{(2)}^{n_2}, \dots, X_{(p)}^{n_p})$$

exist for all finite n_i and

$$E(X_{(2)}^{m_2}, \dots, X_{(p)}^{m_p})$$

for all finite m_j . If the regression equation is

$$(4.1) \quad R(x_{(2)}, \dots, x_{(p)}) E(X_{(1)} | X_{(2)} = x_{(2)}, \dots, X_{(p)} = x_{(p)}) = Q(x_{(2)}, \dots, x_{(p)})$$

then a necessary and sufficient condition that the scedastic equation is given by

$$V(X_{(1)} | X_{(2)} = x_{(2)}, \dots, X_{(p)} = x_{(p)})$$

$$\equiv E[(X_{(1)} - E(X_{(1)} | X_{(2)} = x_{(2)}, \dots, X_{(p)} = x_{(p)}))(X_{(1)} - E(X_{(1)} | X_{(2)} = x_{(2)}, \dots, X_{(p)} = x_{(p)}))']$$

$$(4.2) \quad \begin{aligned} &|X_{(2)} = x_{(2)}, \dots, X_{(p)} = x_{(p)}| \\ &= L(x_{(2)}, \dots, x_{(p)}) \end{aligned}$$

where Q , R and L are square polynomials in $x_{(2)}, x_{(3)}, \dots, x_{(p)}$ and of the same order as $X_{(1)}, X_{(2)}, \dots, X_{(p)}$, follows as

$$(4.6) \begin{pmatrix} a^{(1)} \\ a^{(3)} \end{pmatrix}' M = \begin{pmatrix} \left(\frac{\partial}{\partial a^{(1)}}\right)' \\ \left(\frac{\partial}{\partial a^{(3)}}\right)' \end{pmatrix} M A_{11} + \begin{pmatrix} \left(\frac{\partial}{\partial a^{(2)}}\right)' \\ \left(\frac{\partial}{\partial a^{(4)}}\right)' \end{pmatrix} M A_{21}$$

where $M \equiv M(a_{(1)}, a_{(2)}) = M(a^{(1)}, a^{(2)}, a^{(3)}, a^{(4)})$

Putting $a^{(1)} = a^{(2)} = 0$, it follows that

$$E(X_{(1)} | X_{(2)} = x_{(2)}) = -A_{11}^{-1} A_{12} x_{(2)}$$

Differentiating (4.6) with respect to $(a^{(1)}, a^{(3)})$ it follows that

$$\begin{aligned} \left(\frac{\partial}{\partial a_{(1)}} - \frac{\partial}{\partial a_{(3)}}\right) M \begin{pmatrix} a^{(1)} \\ a^{(2)} \end{pmatrix}' + 2IM = \left(\frac{\partial}{\partial a^{(1)}} - \frac{\partial}{\partial a^{(3)}}\right) \begin{pmatrix} \left(\frac{\partial}{\partial a^{(1)}}\right)' \\ \left(\frac{\partial}{\partial a^{(3)}}\right)' \end{pmatrix} M A_{11} \\ + \left(\frac{\partial}{\partial a^{(1)}} - \frac{\partial}{\partial a^{(3)}}\right) \begin{pmatrix} \left(\frac{\partial}{\partial a^{(2)}}\right)' \\ \left(\frac{\partial}{\partial a^{(4)}}\right)' \end{pmatrix} M A_{12} \end{aligned}$$

Substituting for $\begin{pmatrix} \left(\frac{\partial}{\partial a^{(1)}}\right)' \\ \left(\frac{\partial}{\partial a^{(3)}}\right)' \end{pmatrix}$ from (4.6) and putting $a^{(1)} = a^{(3)} = 0$ the above equation becomes

$$\begin{aligned} \left[\left(\frac{\partial}{\partial a^{(1)}} - \frac{\partial}{\partial a^{(3)}}\right) \begin{pmatrix} \left(\frac{\partial}{\partial a^{(1)}}\right)' \\ \left(\frac{\partial}{\partial a^{(3)}}\right)' \end{pmatrix} M (A_{11}^{-1} A_{12} \left(\frac{\partial}{\partial a^{(2)}} - \frac{\partial}{\partial a^{(4)}}\right)) \right. \\ \left. + (A_{11}^{-1} A_{12} \left(\frac{\partial}{\partial a^{(2)}} - \frac{\partial}{\partial a^{(4)}}\right))' M = 2A_{11}^{-1} M \right]_{a^{(1)} = a^{(3)} = 0} \end{aligned}$$

or

$$\left[\left(\frac{\partial}{\partial a_{(1)}} - \frac{\partial}{\partial a_{(3)}}\right)' M (A_{11}^{-1} A_{12} \frac{\partial}{\partial a_{(2)}}) (A_{11}^{-1} A_{12} \frac{\partial}{\partial a_{(2)}})' M = 2A_{11}^{-1} M \right]_{a_{(1)} = 0}$$

so that by theorem 4.1,

$$V(X_{(1)} | X_{(2)} = x_{(2)}) = 2A_{11}^{-1}.$$

5. Distributions with prescribed Regression Equations and Stochastic Equations

Steyn (1968) used the necessary and sufficient condition (2.1) to derive systems of probability distributions having linear regression equations. The same procedure may be

used to derive systems of probability functions with prescribed regression equations and prescribed scedastic equations. For example, if in the bivariate case the regression equation of X_1 on X_2 and of X_2 on X_1 must be linear and the distribution must further be homoscedastic, it follows from theorems 2.1 and 2.2 without the condition $a_2 = 0$ and $a_1 = 0$ that:

$$(5.1) \quad \frac{\partial}{\partial a_1} M = (a_{11} + a_{12} \frac{\partial}{\partial a_2}) M,$$

$$(5.2) \quad \frac{\partial}{\partial a_2} M = (a_{21} + a_{22} \frac{\partial}{\partial a_1}) M \text{ and}$$

$$(5.3) \quad \frac{\partial^2}{\partial a_1^2} M - (a_{11} + a_{12} \frac{\partial}{\partial a_2})^2 M = b_1 M,$$

$$(5.4) \quad \frac{\partial^2}{\partial a_2^2} M - (a_{21} + a_{22} \frac{\partial}{\partial a_1})^2 M = b_2 M$$

where a_{ij} and b_i are constants.

Clearly, equations (5.1) and (5.2) will lead to $b_1 = b_2 = 0$ i.e. a probability of one associated with a single point on the X_1, X_2 plane. Further it is easily seen by inspection that adding a term $b_1 a_1 M$ to (5.1) and $b_2 a_2 M$ to (5.2) will yield equations (5.3) and (5.4) when $a_1 = a_2 = 0$ and hence give rise to a homoscedastic distribution. It follows by integrating the new equations obtained by adding $b_1 a_1 M$ on $b_2 a_2 M$ respectively that the probability function is then the bivariate normal probability function.

It can easily be shown that further sets of equations may be written down to yield linear regression and homoscedasticity. For example, the equations

$$\frac{\partial}{\partial a_1} M = (a_{11} + a_{12} \frac{\partial}{\partial a_2} + b_1 a_1 + a_1^2 \frac{\partial}{\partial a_2}) M$$

$$\frac{\partial}{\partial a_2} M = (a_{21} + a_{22} \frac{\partial}{\partial a_1} + b_2 a_2 + a_2^2 \frac{\partial}{\partial a_1}) M$$

will also lead to equations (5.1) to (5.4) for $a_1 = 0$ and $a_2 = 0$ i.e. linear regression and homoscedasticity.

It is thus clear that linear regression and homoscedasticity is not sufficient to characterize a bivariate normal distribution.

Similarly, starting with the necessary and sufficient condition for linear regression equations as well as linear scedastic equations i.e. starting with equations (5.1) and (5.2) together with

$$(5.5) \quad \frac{\partial^2}{\partial a_1^2} M - (a_{11} + a_{12} \frac{\partial}{\partial a_2})^2 M = (b_{11} + b_{12} \frac{\partial}{\partial a_2}) M$$

$$(5.6) \quad \frac{\partial^2}{\partial a_2^2} M - (a_{21} + a_{22} \frac{\partial}{\partial a_1})^2 M = (b_{21} + b_{22} \frac{\partial}{\partial a_1}) M$$

it easily follows by inspection that the distribution defined by

$$\frac{\partial}{\partial a_1} M = (a_{11} + a_{12} \frac{\partial}{\partial a_2} + b_{11} a_1 + b_{12} a_1 \frac{\partial}{\partial a_2}) M$$

$$\frac{\partial}{\partial a_2} M = (a_{21} + a_{22} \frac{\partial}{\partial a_1} + b_{21} a_2 + b_{22} a_2 \frac{\partial}{\partial a_1}) M$$

will have the required properties. Again further sets of differential equations leading to other distributions with linear regression equations and linear scedastic equations may be written down.

One can, however, come nearer to the property of uniqueness of distributions with prescribed regression and scedasticity. Steyn (1968) indicated that the set of equations

$$\begin{aligned} \frac{\partial}{\partial a_1} M = & (a_{11} + a_{12} \frac{\partial}{\partial a_2} + a a_1 + k a_1 \frac{\partial}{\partial a_1} + h a_1 \frac{\partial}{\partial a_2} + p a_1 \frac{\partial^2}{\partial a_1^2} + q a_1 \frac{\partial^2}{\partial a_1 \partial a_2} \\ & + r a_1 \frac{\partial^2}{\partial a_2^2}) M \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial a_2} M = & (a_{21} + a_{22} \frac{\partial}{\partial a_1} + b a_2 + l a_2 \frac{\partial}{\partial a_1} + m a_2 \frac{\partial}{\partial a_2} + s a_2 \frac{\partial^2}{\partial a_1^2} + t a_2 \frac{\partial^2}{\partial a_1 \partial a_2} \\ & + v a_2 \frac{\partial^2}{\partial a_2^2}) M \end{aligned}$$

leads to the bivariate Pearson system with linear regression equations. Within this system the equations (5.1) to (5.4) and (5.1), (5.2), (5.5), (5.6) uniquely determines the respective distributions.

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